INDUSTRIAL INTERNSHIP AT BHEL TIRUCHIRAPPALLI

A SUMMER INTERNSHIP REPORT

Submitted by

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BONAFIDE CERTIFICATE

This is to certify this Summer Internship project report **"Industrial Internship at BHEL Tiruchirappalli"** is the Bonafide work of **ANUSUYA V (810023205032)** has satisfactory completed the course of **IT3711 – SUMMER INTERNSHIP** for **SEVENTH** semester, **B.TECH INFORMATION TECHNOLOGY** during the academic year **2025-2026.**

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ABSTRACT

This report documents the summer internship undertaken in the Digital Transformation Group (DTG) of Bharat Heavy Electricals Limited (BHEL), Tiruchirappalli, the unit-level IT organisation that plans, operates and secures the digital infrastructure of the High Pressure Boiler Plant (HPBP) and the Seamless Steel Tube Plant (SSTP). The internship was confined to enterprise information technology.

The campus network follows a hierarchical core–distribution–access design with IEEE 802.1Q VLAN segmentation and a clustered Check Point next-generation firewall at the perimeter. The captive data centre hosts the SAP HANA database on IBM Power Systems servers partitioned by PowerVM, with block storage on a Fibre Channel SAN, file shares on NAS, and protection through high availability, disaster recovery, RAID 6 and 3-2-1 backups. SAP S/4HANA runs on a three-tier architecture with changes moved through a DEV–QAS–PRD transport landscape, and its MM, PP, QM, PM, SD and FICO modules post into a single Universal Journal.

Security follows a defence-in-depth model in which an IT–OT demilitarised zone isolates plant control systems, Harmony Endpoint protects user devices and ThreatCloud AI shares threat intelligence. Operations follow ITIL 4 practices for incidents, problems and changes, recorded in ServiceNow with P1–P4 priorities and SLAs.

The report presents the company profile, a thirty-day work log, architecture diagrams, algorithms, scripts developed during the internship, text-based sample screens, an analysis of security and storage design, and the learning outcomes and future scope of the internship.

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CHAPTER 1
INTRODUCTION

1.1 Objective of the Internship

The first objective of this internship was to understand how the information technology concepts taught in the B.Tech curriculum behave at enterprise scale. In a college laboratory, a network is usually a single flat subnet, every student holds administrator rights and a misconfiguration costs only a repeated experiment. At BHEL Tiruchirappalli, the same subnet, Linux server and database support thousands of business transactions every shift, so availability, auditability and security are primary design requirements rather than optional extras.

The second objective was to gain exposure to enterprise platforms that are rarely available in college, namely IBM Power Systems with PowerVM, SAN storage, SAP S/4HANA on the HANA database, Check Point firewalls and ITSM tools, and to learn the process discipline of the ITIL 4 framework, in which every change is requested, approved, scheduled, implemented with a rollback plan and reviewed.

1.2 Motivation

Manufacturing is moving through Industry 4.0, whose defining feature is the convergence of Information Technology (IT), such as ERP, directory services and the data centre, with Operational Technology (OT), the systems that run the production floor. Convergence replaces manual end-of-shift reconciliation with near-real-time data flows, which turns the enterprise network and the ERP system into the nervous system of the factory.

Convergence also creates the most important security challenge in modern manufacturing, because OT systems prioritise availability and safety and often use protocols without authentication. The accepted solution is a demilitarised zone with firewalls on both sides. At BHEL Tiruchirappalli, DTG owns this IT boundary, which made the group an ideal place to study IT–OT convergence, SAP and enterprise security from the IT side.

CHAPTER 2
COMPANY PROFILE

2.1 BHEL Tiruchirappalli (HPBP and SSTP)

Bharat Heavy Electricals Limited (BHEL) is India's largest engineering and manufacturing enterprise in the energy and infrastructure sector and a Government of India undertaking under the Ministry of Heavy Industries. Incorporated on 13 November 1964 and conferred Maharatna status in 2013, it operates sixteen manufacturing divisions, and BHEL-supplied equipment accounts for roughly 53 percent of India's installed utility-scale thermal, hydro and nuclear generation capacity.

BHEL Tiruchirappalli was the first plant commissioned after incorporation and began production in May 1965. Its campus has two units: the High Pressure Boiler Plant (HPBP), one of the largest boiler-manufacturing facilities in the world, and the Seamless Steel Tube Plant (SSTP), which supplies tubes to HPBP and external customers. Both are project-based manufacturers whose multi-year orders generate thousands of purchase orders, production orders and inspection lots, and both share one campus network, one data centre and one SAP landscape operated by DTG.

2.2 Digital Transformation Group (DTG)

The Digital Transformation Group is the unit-level IT organisation of BHEL Tiruchirappalli. It reports to the IT and Digital Transformation head, who reports to the Executive Director, and it coordinates with BHEL's corporate IT organisation for policies, security standards and major programmes such as the SAP S/4HANA rollout. Table 2.1 summarises its five functional teams.

Team	Responsibility	Key Platforms
Infrastructure and Data Centre	Servers, virtualisation, storage, backup	IBM Power, PowerVM, FlashSystem SAN, NAS
Network	LAN/WAN design, VLANs, routing, QoS	Core, distribution and access switches, WAN routers
Applications and ERP	SAP Basis, transports, module support	SAP S/4HANA, SAP GUI, Fiori
Security	Firewalls, endpoint protection, access control	Check Point Quantum, Harmony Endpoint
End-User Support	L1–L2–L3 helpdesk, tickets, SLAs	ServiceNow, Nagios, PRTG

Table 2.1 Functional Teams of DTG

A typical DTG engineer's day begins with a review of overnight monitoring alerts and backup logs, continues through ticket resolution, change coordination and project work, and closes with documentation and shift handover, which is the rhythm followed in the work log of Chapter 3.

CHAPTER 3
DAILY WORK REPORT

The thirty-day internship was organised into six weekly themes. All hands-on work was carried out on read-only consoles, sandbox systems and training machines, and no change was made to any production system.

3.1 Week 1 – Enterprise Networking Fundamentals (Days 1–5)

Day 1 – Induction, IT Security Briefing and DTG Orientation

On the first day, I reported to the training section of BHEL Tiruchirappalli for induction, where my documents were verified, a temporary access badge was issued and I signed the organisation's IT Acceptable Use Policy and confidentiality undertaking. I was then introduced to my mentor in the Digital Transformation Group, who explained the five functional teams of the group, namely Infrastructure and Data Centre, Network, Applications and ERP, Security, and End-User Support. My mentor made it clear that interns work only on read-only consoles, sandbox systems and documentation, and that no change may ever be made to a production system without an approved change ticket.

Day 2 – Study of the Campus Network Topology

I studied the sanitised network topology documentation of the BHEL Trichy campus, which follows the hierarchical core–distribution–access model used in large enterprise networks. The core layer consists of a redundant pair of Layer-3 switches joined by an aggregated inter-switch link, the distribution layer aggregates buildings such as the HPBP and SSTP office blocks and the data centre server farm, and the access layer connects end devices such as PCs, IP phones and data-capture terminals. Accompanied by a network engineer, I visited a building distribution room and observed the patch panels, fibre uplinks, rack labelling scheme and the dual uplinks that connect each distribution switch to both core switches.

Day 3 – IP Addressing, Subnetting and DHCP Scope Planning

The third day was devoted to IP addressing, where my mentor explained how the campus uses private IPv4 ranges with a separate subnet for each department, VLAN and server segment. As an exercise, I designed a Variable Length Subnet Mask plan for a hypothetical 10.20.0.0/16 block, allocating a /22 for office users, /24 subnets for voice

and printers, /26 subnets for server farms and /30 point-to-point links for router interconnections. I then compared my plan with the documented addressing conventions and understood why servers, network devices and firewalls receive static addresses while end-user devices obtain addresses dynamically.

Day 4 – VLAN Segmentation and 802.1Q Trunking

I studied how VLANs separate office data, voice, plant data-capture terminals, printers and network management into independent broadcast domains on shared switching hardware. Under supervision, I observed the output of read-only commands such as show vlan brief and show interfaces trunk on an access switch, which listed the access ports assigned to each VLAN and the tagged VLANs allowed on the uplink trunk. My mentor explained the voice VLAN feature, through which an IP phone and the PC connected behind it are placed in different VLANs on the same physical port so that voice traffic can be prioritised.

Day 5 – Inter-VLAN Routing, WAN Connectivity and QoS

On the fifth day, I learned how traffic moves between VLANs through Switched Virtual Interfaces on the core Layer-3 switches, which act as default gateways for each subnet and are protected by a first-hop redundancy protocol. I then traced the path of traffic leaving the campus, from the core switches through the perimeter firewall to the WAN routers that connect BHEL Trichy with Bhopal, Haridwar, Hyderabad and the corporate office in New Delhi. Using ping and traceroute from my test workstation, I measured round-trip latency to an internal corporate portal and observed the hops between the campus and the WAN.

3.2 Week 2 – Directory Services and Linux Administration (Days 6–10)

Day 6 – DNS and DHCP Services in Practice

I studied the DNS and DHCP infrastructure, which DTG classifies as utility services because an outage of either can make every other application unreachable even when the application servers are healthy. Using nslookup on my test workstation, I queried forward and reverse lookup zones of the internal domain and observed how internal names resolve to static server addresses. With permission, I captured a DHCP exchange in Wireshark on an isolated test PC and identified the four DORA messages

of Discover, Offer, Request and Acknowledge together with the options for subnet mask, default gateway and DNS server.

Day 7 – Active Directory, Group Policy and User Lifecycle

I was introduced to Active Directory, which provides centralised identity, Kerberos authentication and authorisation for the Windows domain, with at least two domain controllers for redundancy. In a read-only view of Active Directory Users and Computers, I studied the Organisational Unit structure arranged by department and observed how security groups, not individual users, are granted access to file shares and applications. I observed an L2 engineer process an approved ticket for a new employee account, add it to role-based groups, and separately disable the account of a transferred employee on the same day as required by policy.

Day 8 – Linux Server Administration through PuTTY and SSH

On this day, I worked on a test Red Hat Enterprise Linux virtual machine provided for training, since the SAP HANA database tier at BHEL Trichy runs on Linux. I connected through PuTTY using SSH on TCP port 22, generated an RSA key pair and configured public-key authentication, after which my mentor explained why password authentication is disabled for administrative access on production servers. I practised core commands such as ls, cd, grep, ps, top, df, free, chmod and chown, and used systemctl to check, stop, start and enable a sample service.

Day 9 – Writing a Bash Server Health-Check Script

I was assigned to write a Bash script that automates the routine morning health check performed on Linux servers. The script collects the hostname, uptime and load average, memory usage from free, and file-system utilisation from df, and compares each mounted file system against warning and critical thresholds of 80 and 90 percent. Results are written with a timestamp to a log file, and any file system above the critical threshold is highlighted at the top of the report so that an engineer can act on it immediately.

Day 10 – Log Analysis on Linux and Windows

I studied log analysis as the primary diagnostic and security-investigation tool available to administrators. On the Linux test server, I used journalctl to filter service logs by unit and time, tail -f to follow logs live, and grep on /var/log/secure to identify failed SSH login attempts that I generated deliberately from another machine. I

extended my Day 9 script with an awk pipeline that counts failed logins per source IP address and appends the top offenders to the daily report.

3.3 Week 3 – Data Centre, PowerVM and Storage (Days 11–15)

Day 11 – Data Centre Facility Tour

Escorted by the infrastructure team, I visited the BHEL Trichy data centre after my visit was recorded in the access register, which demonstrated that physical access control is the first layer of security. I observed the power chain from utility supply through the UPS systems to rack-level power distribution units, with diesel generators providing backup during extended outages. The engineer explained the precision air-conditioning units, the hot-aisle and cold-aisle arrangement of racks, the fire-detection and suppression system, and the environmental sensors that alert the team to temperature or humidity changes.

Day 12 – IBM Power Systems and PowerVM Fundamentals

I studied IBM Power Systems, the enterprise servers that host the SAP HANA database tier, and their reliability, availability and serviceability features such as error-correcting memory, hot-swappable components and processor-level fault isolation. My mentor explained the PowerVM hypervisor, which runs in firmware and divides a physical server into Logical Partitions, each with its own operating system and allocated processor and memory resources. I learned the difference between dedicated processors and shared processor pools, and how entitled capacity and virtual processors allow an uncapped partition to borrow idle cycles from the pool.

Day 13 – Hardware Management Console and Live Partition Mobility

I observed, in read-only mode, the Hardware Management Console used to manage the Power Systems servers and their partitions. The console showed each managed system with its partitions, their running state, processor entitlement and memory allocation, and the engineer pointed out how the SAP HANA partition is sized with dedicated resources because in-memory databases are highly sensitive to memory capacity. My mentor explained Live Partition Mobility, which moves a running partition to another physical server without downtime, allowing firmware updates and hardware maintenance during working hours without interrupting SAP.

Day 14 – SAN Storage, Fibre Channel Zoning and RAID

I studied the storage architecture in which IBM FlashSystem dual-controller arrays provide block storage to servers over a dedicated Fibre Channel Storage Area Network. My mentor explained that each server Host Bus Adapter port has a World Wide Port Name, and that zoning on the SAN switches ensures a server can see only the storage ports and volumes assigned to it. On the test Linux system, I examined multipath output that showed several paths to the same volume, which explained how the loss of one HBA, cable or switch does not interrupt access to the database volumes.

Day 15 – Backup Policy, High Availability and Disaster Recovery

I studied the data-protection strategy, which treats RAID, backup and disaster recovery as complementary layers protecting against disk failure, logical data loss and loss of an entire site respectively. My mentor explained the 3-2-1 backup principle followed for SAP data and compared full, incremental and differential backups in terms of backup time and restore effort. I reviewed the previous night's backup job reports with the engineer, learned to read job status, data volume and duration, and noted that a single failed job is raised as an incident ticket rather than being ignored.

3.4 Week 4 – SAP S/4HANA Architecture and Basis (Days 16–20)

Day 16 – SAP S/4HANA Three-Tier Architecture

I began the SAP week by studying the three-tier architecture of SAP S/4HANA, consisting of the presentation tier, the application tier and the database tier. Using a training system, I logged on through SAP GUI and through the Fiori Launchpad in a web browser, and understood that both are thin presentation layers while the business logic executes on the application servers. My mentor explained the ABAP Central Services instance, which provides the message server and the enqueue server, and the primary and additional application server instances that host the work processes.

Day 17 – SAP Landscape and Transport Management

On this day, I studied the three-system SAP landscape of Development, Quality Assurance and Production, and the rule that no configuration or code change is ever made directly in Production. In the development system, my mentor showed me a transport request in the Transport Organizer through transactions SE09 and SE10, explaining how changes are recorded under tasks and how the request is released after unit testing. I then observed in the Transport Management System, transaction STMS,

how the released request appeared in the import queue of the quality system and was imported after the change ticket received approval.

Day 18 – SAP Basis Monitoring

I shadowed an SAP Basis administrator during the daily system health check performed on the SAP landscape. Using transaction ST22, we reviewed ABAP runtime errors from the previous day, and with SM21 we examined the system log for warnings related to work processes, memory and database connectivity. In SM37, I learned to filter background jobs by status, identify cancelled jobs and read job logs to find the cause of a failure.

Day 19 – SAP User Administration and Authorisations

I studied SAP user and authorisation management, which is controlled strictly because SAP holds the commercial and financial data of the organisation. In a training client, I observed user creation in SU01 and role maintenance in PFCG, where a role groups transactions and authorisation objects that are then assigned to users through approved tickets. My mentor explained the principle of segregation of duties, under which the same user should not, for example, both create a purchase order and approve the related vendor invoice.

Day 20 – SAP Functional Modules and Integration

I spent the day exploring the core functional modules used at BHEL Trichy in a sandbox client so that no live business data was affected. In Materials Management, I created a purchase order with ME21N, posted a goods receipt with MIGO and checked the resulting stock in MMBE, and I observed how the goods receipt automatically created a financial document in FICO. In Production Planning, I displayed a production order with CO03 and studied its bill of materials, routing and operations, and I learned how Quality Management inspection lots can block stock until a usage decision is recorded.

3.5 Week 5 – Cybersecurity and Firewall Operations (Days 21–25)

Day 21 – Security Policies and the Defence-in-Depth Model

I began the security week by studying the IT security policies of the organisation, including the Acceptable Use Policy, password and multi-factor authentication requirements, data classification rules and incident-reporting obligations. My mentor explained the defence-in-depth model followed at DTG, in

which network security, platform hardening, storage protection and endpoint security each provide an independent layer around critical assets. We discussed the CIA triad of confidentiality, integrity and availability, and why the order of priority shifts towards availability and safety on the OT side of the plant.

Day 22 – Review of Check Point Firewall Rule Base

I was given read-only access to Check Point SmartConsole to study the rule base of the Quantum Security Gateway cluster that protects the network perimeter. My mentor explained that rules are evaluated from top to bottom with the first match applied, which is why the stealth rule protecting the gateway itself appears near the top and the cleanup rule that drops and logs all remaining traffic appears at the end. I reviewed a section of the rule base against a checklist and identified rules with very broad sources or services, rules with zero hit counts over the past ninety days, and rules missing descriptive comments or ticket references.

Day 23 – IT–OT DMZ and Boundary Firewall Policy

I studied the architecture of the demilitarised zone that separates the enterprise IT network from the operational technology network of the plant. My mentor explained that no session is permitted to pass directly from IT to OT, and that traffic must terminate on a server in the DMZ, such as a data replication server, a patch and antivirus update relay, a secure file-transfer server or a jump host for authorised remote access. In SmartConsole, I observed how the rules on the IT-facing side and the OT-facing side are written as narrowly defined permits between specific hosts and ports, with all other traffic denied and logged.

Day 24 – Endpoint Protection with Harmony Endpoint

I studied Check Point Harmony Endpoint, the endpoint protection platform installed on the organisation's desktops and laptops. Through a read-only console view, I observed the policy blades for anti-malware, anti-ransomware, threat emulation, URL filtering and zero-phishing protection, and the compliance status of endpoints that were missing updates. My mentor walked me through the forensic report of a previously blocked malicious attachment, which showed the attack chain from the e-mail download to the blocked execution of the file.

Day 25 – Patch Management and Vulnerability Prioritisation

I studied the patch management process, which DTG treats as a core security control because most successful intrusions exploit vulnerabilities for which a patch already exists. My mentor explained the monthly cycle for Windows systems through WSUS and the package-manager-based cycle for Linux systems, both applied in scheduled maintenance windows outside production shifts after testing on non-production servers. Using a vulnerability report from a test segment, I learned to prioritise findings by CVSS score, exposure of the system to the internet or the DMZ, and the business criticality of the asset.

3.6 Week 6 – ITIL Service Management and Consolidation (Days 26–30)

Day 26 – ITIL 4 Practices and Helpdesk Shadowing

I studied the ITIL 4 framework and its four practices most used by DTG, namely incident management, problem management, service request management and change enablement. I then shadowed the L1 helpdesk for the day and observed how calls and e-mails are converted into tickets with a category, a configuration item, a priority and an assignment group. Most tickets were routine service requests such as password resets, printer mapping and SAP access requests, which follow pre-approved fulfilment procedures, while a smaller number were incidents that required troubleshooting.

Day 27 – Change Management and the Change Advisory Board

I attended a Change Advisory Board meeting as an observer, where planned changes for the coming week were reviewed by representatives of the network, server, SAP and security teams. Each Request for Change described the change, the business justification, the affected configuration items, the implementation window, the rollback plan and the verification steps. I observed how one change, an operating-system patch on a group of application servers, was approved with a condition to complete it outside the main production shift, while another was deferred because its rollback plan was incomplete.

Day 28 – Infrastructure Monitoring with Nagios and PRTG

I studied the monitoring platforms used by DTG, Nagios for flexible open-source host and service checks and PRTG for sensor-based monitoring with visual dashboards. On a training Nagios instance, I observed host and service definitions and

checks such as check_ping, check_disk and check_load, together with their warning and critical thresholds. I then converted my Day 9 health-check script into a Nagios-compatible plugin that prints a single status line with performance data and returns exit code 0 for OK, 1 for WARNING, 2 for CRITICAL and 3 for UNKNOWN.

Day 29 – Problem Management and Knowledge Documentation

I studied problem management, which seeks the root cause of recurring or significant incidents rather than only restoring service. My mentor walked me through a closed problem record in which repeated disk-space alerts on a log partition were traced to a log-rotation configuration that had not been applied after an application update. Using the five-whys technique, we reconstructed how the incidents had been resolved temporarily by clearing old files until the problem record identified the configuration gap and raised a change to correct it.

Day 30 – Consolidation, Presentation and Exit Formalities

On the final day, I consolidated my daily notes, scripts, diagrams and checklists into a structured internship record and reviewed it with my mentor for technical accuracy and for the removal of any sensitive information. I presented a summary of my learning to the DTG team, covering the network hierarchy, the PowerVM and SAN architecture, the SAP landscape and transport process, the Check Point rule-base review and the ITIL workflow. The engineers provided feedback on my presentation, particularly on explaining the IT–OT boundary in terms that a non-specialist manager could understand.

CHAPTER 4
BLOCK DIAGRAMS

The following text-based block diagrams represent the three principal architectures studied. Device names and numbers are illustrative and do not represent the actual BHEL configuration.

4.1 Enterprise Core-Distribution-Access Network Architecture

The core layer provides redundant Layer-3 routing towards the WAN and internet edge, the distribution layer aggregates buildings and terminates VLANs, and the access layer connects end devices. Every distribution switch has links to both core switches, so there is no single point of failure.

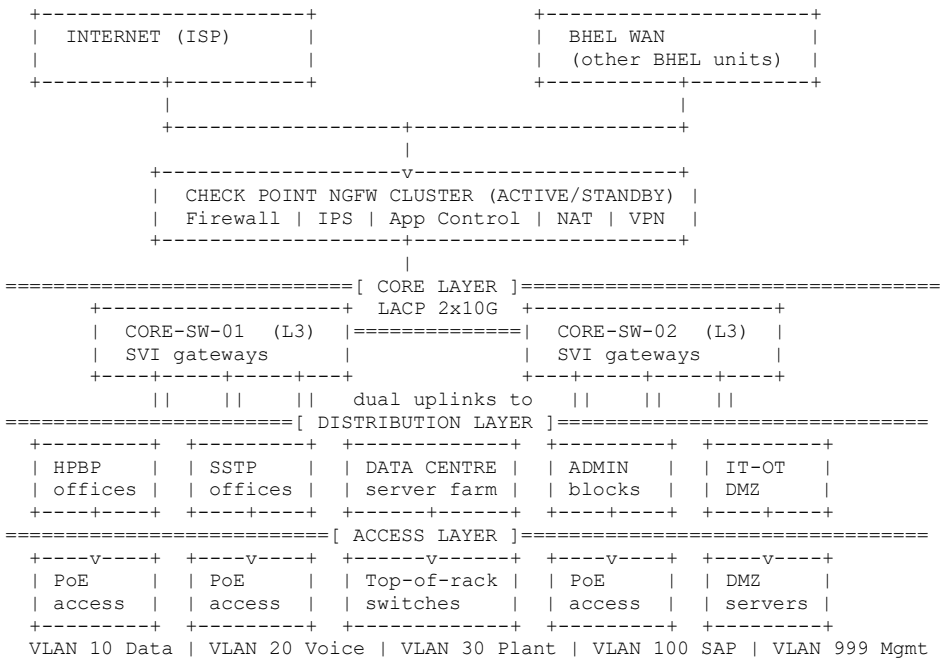


Fig. 4.1 Core-Distribution-Access Network Architecture (illustrative)

Layer	Components	Function in the BHEL Trichy network
Edge	Internet edge router, WAN router	Terminate ISP and inter-unit WAN links; apply edge ACLs and WAN QoS so that voice and SAP traffic receive priority over bulk transfers.

Layer	Components	Function in the BHEL Trichy network
Perimeter security	Check Point Quantum NGFW cluster (active/standby)	Stateful inspection, intrusion prevention, application control, NAT and VPN between the campus and all external networks, with state synchronisation for seamless failover.
Core	Two Layer-3 core switches joined by an LACP port-channel	High-speed routing between distribution blocks, SVI default gateways with first-hop redundancy, DHCP relay towards central servers.
Distribution	Per-block distribution switches (HPBP, SSTP, data centre, administration, DMZ)	Aggregate access switches, terminate VLANs, enforce inter-VLAN policy and provide dual uplinks to both core switches.
Access	48-port PoE access switches and top-of-rack server switches	Connect PCs, IP phones, printers, data-capture terminals and servers; enforce port security, voice VLAN and spanning-tree protection.
Logical overlay	802.1Q VLANs and per-VLAN subnets	Separate office, voice, plant data-capture, server, backup, DMZ and management traffic into isolated broadcast and security domains.

Table 4.1 Structural Description of the Core-Distribution-Access Architecture

4.2 IT-OT Demilitarized Zone (DMZ) Architecture

The DMZ follows the zoned approach of the Purdue reference model: the enterprise zone and the operational zone never communicate directly, and every permitted exchange terminates on a server in the intermediate zone protected by firewalls on both sides.

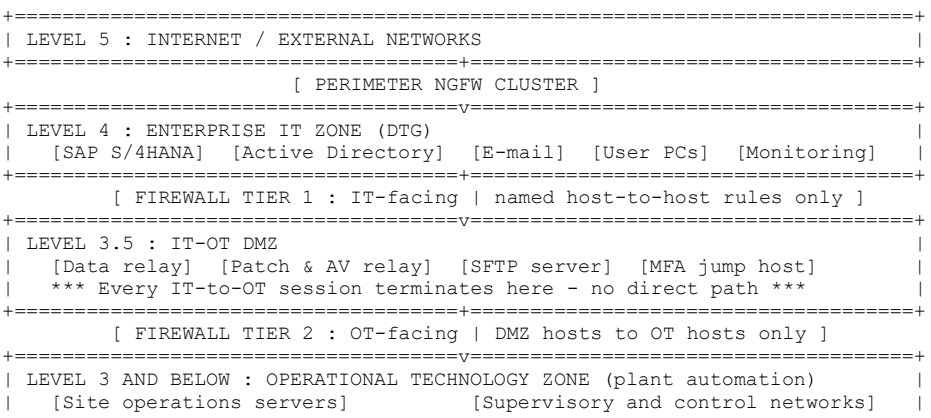


Fig. 4.2 IT-OT DMZ Architecture (illustrative)

4.3 SAP S/4HANA 3-Tier Architecture

SAP S/4HANA separates the user interface, business logic and data into three tiers that scale independently, and every change reaches production only through the DEV–QAS–PRD transport route.

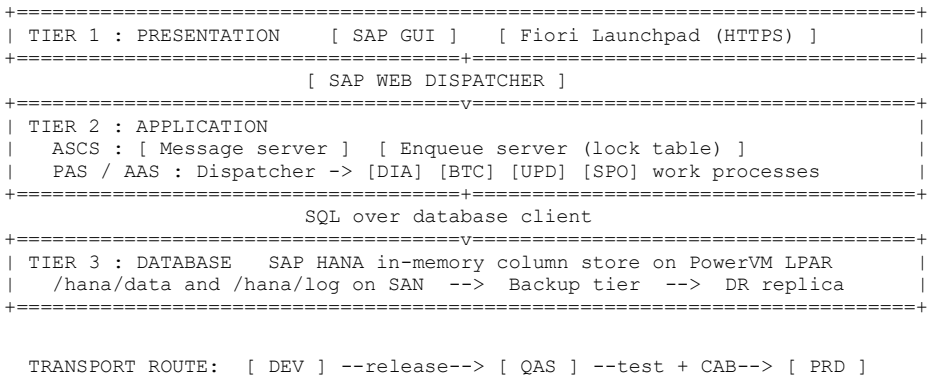


Fig. 4.3 SAP S/4HANA Three-Tier Architecture and Landscape (illustrative)

CHAPTER 5
BHEL ENTERPRISE IT SYSTEM

5.1 Existing System

In the legacy model, manufacturing organisations ran separate departmental applications for materials, production, quality, maintenance and finance, each with its own copy of master data and connected only by nightly file transfers or manual re-entry. Stock was reconciled only at the end of a shift or month, costs of long-cycle contracts were known only after completion, and departmental reports routinely disagreed. Servers were bought department by department without redundant power or tested backups, each kept its own user accounts, and flat networks allowed a single infected PC to affect a whole building.

5.2 Proposed System / Enterprise Architecture

The consolidated architecture is built around a captive data centre engineered with N+1 UPS and cooling, dual power feeds to every rack, diesel-generator backup and hot- and cold-aisle containment. Critical workloads, above all SAP HANA, run on IBM Power Systems: Power10 provides up to 15 SMT8 cores per chip with transparent memory encryption, while Power11 adds spare cores and fewer planned outages for firmware updates. PowerVM partitions each server into Logical Partitions, with dedicated processors for production HANA.

Storage uses two complementary architectures, compared in Table 5.1. The SAN presents block storage from IBM FlashSystem arrays over two separate Fibre Channel fabrics, while NAS exports shared file systems over the IP network.

Attribute	SAN (Block Storage)	NAS (File Storage)
Transport	Dual Fibre Channel fabrics, 32/64 Gbps	Standard Ethernet / IP
Protocols	SCSI or NVMe over FC	NFS v4, SMB 3
Sharing	One host or cluster owns each volume	Many clients share files concurrently
Access control	Zoning by WWPN, host mapping	Export rules, ACLs, AD integration
Typical use	HANA data and log volumes	Department shares, /usr/sap/trans

Table 5.1 Comparison of SAN and NAS

5.3 Capabilities

The infrastructure provides layered protection, summarised in Table 5.2. High availability uses HANA System Replication in syncmem mode with a cluster manager that moves the service automatically; disaster recovery replicates asynchronously to a recovery site within agreed RPO and RTO targets; RAID 6 uses dual parity to survive two drive failures with (n−2)/n usable capacity; and the 3-2-1 backup rule keeps three copies on two media with one off-site, with HANA log backups every 15 minutes by default.

Capability	Protects Against	Indicative Recovery
High availability	Server, partition or path failure within the site	Automatic, seconds to minutes, zero data loss
Disaster recovery	Loss of the entire data centre	Per agreed RTO; RPO of seconds to minutes
RAID 5 / RAID 6	One / two simultaneous drive failures	Online rebuild, no service impact
3-2-1 backup	Deletion, corruption, ransomware	Point-in-time restore

Table 5.2 Infrastructure Capabilities

CHAPTER 6
PROPOSED METHOD (IMPLEMENTATION)

6.1 Server Administration

Linux servers running Red Hat Enterprise Linux are administered through systemd (systemctl, journalctl), the Logical Volume Manager with XFS file systems, and tools such as top, vmstat, iostat and sar. Bash scripts follow defensive conventions: set -euo pipefail, quoted variables, timestamped logs and meaningful exit codes, with privileged commands run through narrowly scoped sudo rules.

Active Directory provides Kerberos-based identity for Windows and, through realmd and SSSD, for Linux. Five FSMO roles are assigned to specific domain controllers, clients find controllers through DNS SRV records, Group Policy applies in Local, Site, Domain and OU order, and access follows the AGDLP model, so a role change requires only one group-membership change.

6.2 Network Management

VLANs follow IEEE 802.1Q, which inserts a 4-byte tag containing a 12-bit VLAN ID and a 3-bit priority field. Trunks carry only allowed VLANs with an unused native VLAN to prevent VLAN hopping, and inter-VLAN routing with access lists runs on the core switches. Subnets are sized with VLSM, where a /n prefix gives 2^(32−n) − 2 usable hosts, so a /22 serves 1,022 hosts and a /26 serves 62.

Voice runs on Avaya Aura: System Manager administers users and dial plans, Session Manager routes SIP, Communication Manager processes calls and G450 Media Gateways connect to the public network. RTP media is marked DSCP 46 in a priority queue, targeting under 150 ms one-way delay (ITU-T G.114), and a G.711 call uses about 87 kbps per direction on Ethernet.

6.3 SAP Module Integration

All postings in SAP S/4HANA land in the Universal Journal (ACDOCA) and all stock movements in MATDOC, so finance and logistics read the same line items. MRP Live (MD01N) creates planned orders and purchase requisitions; a goods receipt (MIGO, movement type 101) creates a QM inspection lot that blocks stock until a usage

decision in QA11; and components are issued to production orders with movement type 261.

Account determination (OBYC) posts each movement automatically: BSX for inventory, WRX for goods-received/invoice-received clearing and GBB for consumption. Plant Maintenance orders settle costs to cost centres, and on the sales side a delivery with post goods issue (movement type 601) records cost of goods sold before billing in VF01 posts revenue and the receivable.

6.4 Cybersecurity

Check Point Quantum Security Gateways run as ClusterXL high-availability pairs at the perimeter and the IT–OT boundary, managed centrally through SmartConsole. SecureXL and CoreXL allow inspection at line rate, and software blades provide firewall, IPS, application control, URL filtering, anti-bot, threat emulation (sandboxing) and threat extraction (sanitised documents).

Harmony Endpoint protects user devices with anti-malware, behavioural detection, anti-ransomware with file rollback, zero-phishing and automated forensic reports. ThreatCloud AI shares threat intelligence across all gateways and endpoints, so an indicator first seen elsewhere can be blocked at BHEL Trichy without waiting for a signature update.

6.5 IT Service Management

ITIL 4 organises service management through the Service Value System, seven guiding principles and practices such as incident, problem, change enablement and service request management. ServiceNow implements these on a Configuration Management Database that links incidents and changes to the affected servers and services, routes tickets to assignment groups, and runs SLA timers with escalation.

Priority is derived from impact and urgency, as shown in Table 6.1. A P1 incident triggers the major incident procedure and a mandatory problem record after service is restored.

Impact \ Urgency	High	Medium	Low
High (plant-wide or critical service)	P1 – Critical	P2 – High	P3 – Medium
Medium (department or major function)	P2 – High	P3 – Medium	P4 – Low

Impact \ Urgency	High	Medium	Low
Low (single user, workaround exists)	P3 – Medium	P4 – Low	P4 – Low

Table 6.1 Incident Priority Matrix

CHAPTER 7
ALGORITHMS & WORKFLOWS

7.1 Linux Remote Administration via SSH/PuTTY and Service Restart

Step 1: Confirm that an approved incident or change ticket exists and note its number.

Step 2: Load the passphrase-protected private key into PuTTY (Pageant).

Step 3: Open an SSH session to the server on TCP port 22, through the jump host if required.

Step 4: Verify the host-key fingerprint against the inventory; abort on mismatch.

Step 5: Log on with the named personal account using public-key authentication.

Step 6: Diagnose with systemctl status, journalctl -u, df -h and free -m.

Step 7: Correct the cause, then run sudo systemctl restart and confirm with systemctl is-active.

Step 8: Verify the monitoring check returns to OK, update the ticket and resolve it.

7.2 SIP VoIP Signaling Call Flow (INVITE, RTP, BYE)

Step 1: Both phones REGISTER with Session Manager and receive 200 OK.

Step 2: The caller sends INVITE with an SDP offer (IP address, RTP port, codecs).

Step 3: Session Manager replies 100 Trying and applies the dial plan with Communication Manager.

Step 4: The INVITE is forwarded to the callee, or to a G450 gateway for external numbers.

Step 5: The callee rings and returns 180 Ringing along the Via path.

Step 6: On answer, the callee returns 200 OK with the SDP answer selecting a common codec.

Step 7: The caller sends ACK; RTP media flows over UDP every 20 ms, with RTCP quality reports.

Step 8: Either side sends BYE, the other replies 200 OK, and a Call Detail Record is written.

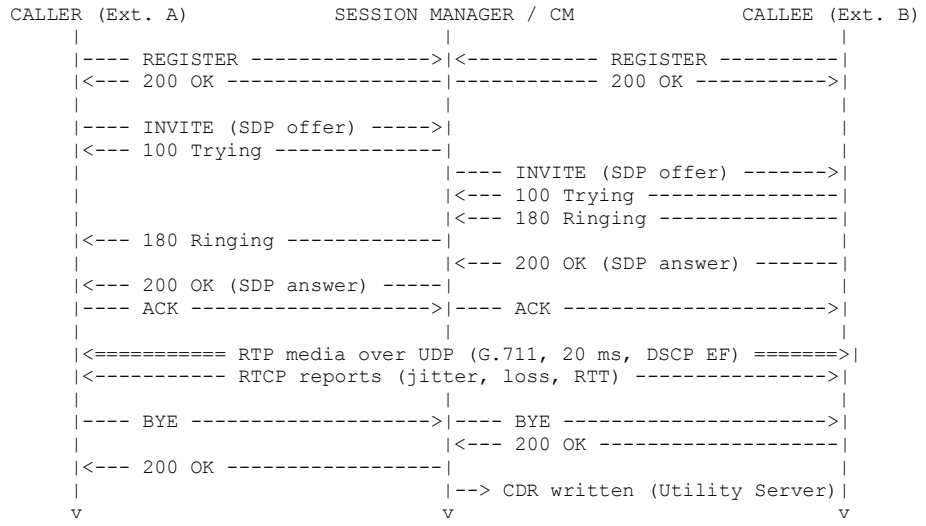


Fig. 7.1 SIP Call Flow Ladder Diagram (signalling via Session Manager, media end-to-end)

7.3 SAP Order-to-Delivery Cross-Module Execution Flow

Step 1: Create the sales order in VA01 (SD) with pricing and availability check.

Step 2: Run MRP Live (MD01N) to create planned orders and purchase requisitions (PP).

Step 3: Convert requisitions to purchase orders (ME21N) and post goods receipt with MIGO 101 (MM, FI).

Step 4: Record the usage decision in QA11 to release stock from quality inspection (QM).

Step 5: Convert and release the production order (CO40/CO02); issue components with 261 (PP, MM).

Step 6: Confirm operations in CO11N and receive finished goods with 101 against the order.

Step 7: Create the delivery (VL01N) and post goods issue with 601 (SD, MM, FI).

Step 8: Bill in VF01 and settle the production order in CO88, posting to ACDOCA (FI, CO).

7.4 ITIL Incident-to-Change Resolution Cycle

Step 1: Log the incident in ServiceNow against the affected configuration item.

Step 2: Set priority from impact and urgency, which starts the SLA timers.

Step 3: Route to the resolver group; check the knowledge base and recent changes.

Step 4: Escalate to L2, L3 or the major incident procedure if not resolved within the SLA threshold.

Step 5: Restore service, confirm with the user and resolve the incident.

Step 6: Open a problem record for P1 or recurring incidents and find the root cause.

Step 7: Raise a Request for Change with risk, rollback plan and window; obtain CAB approval.

Step 8: Implement, verify and hold a post-implementation review; update the KB and CMDB.

CHAPTER 8
CODE IMPLEMENTATION

The scripts below were written and tested only on training systems; names and addresses are illustrative.

8.1 Linux Bash Script for Automated Server Health Checks

The script measures CPU from two /proc/stat samples, memory from MemAvailable and disk usage from df, and returns Nagios-compatible exit codes (0 OK, 1 WARNING, 2 CRITICAL).

```
#!/usr/bin/env bash
# server_health.sh - CPU, memory and disk check with Nagios exit codes
set -euo pipefail
WARN=80; CRIT=90; STATUS=0
LOG="/var/log/health_$(hostname -s)_$(date +%F).log"
raise() { if (( $1 > STATUS )); then STATUS=$1; fi; }
rate()  {
    if   (( $2 >= CRIT )); then echo "CRITICAL $1=$2%" | tee -a "$LOG"; raise 2
    elif (( $2 >= WARN )); then echo "WARNING  $1=$2%" | tee -a "$LOG"; raise 1
    else                        echo "OK       $1=$2%" | tee -a "$LOG"; fi
}
read -r _ u1 n1 s1 i1 w1 x1 y1 z1 _ < /proc/stat; sleep 1
read -r _ u2 n2 s2 i2 w2 x2 y2 z2 _ < /proc/stat
idle=$(( (i2 + w2) - (i1 + w1) ))
total=$(( (u2+n2+s2+i2+w2+x2+y2+z2) - (u1+n1+s1+i1+w1+x1+y1+z1) ))
rate "cpu" $(( 100 * (total - idle) / total ))
mt=$(awk '/^MemTotal:/{print $2}' /proc/meminfo)
ma=$(awk '/^MemAvailable:/{print $2}' /proc/meminfo)
rate "memory" $(( 100 * (mt - ma) / mt ))
while read -r pct mnt; do rate "disk:$mnt" "${pct%\%}"
done < <(df -P -x tmpfs -x devtmpfs | awk 'NR>1 {print $5, $6}')
exit "$STATUS"
```

Listing 8.1 server_health.sh

8.2 Cisco Switch CLI for VLAN and Port-Security Configuration

The configuration creates VLANs, sets user ports as access ports with a voice VLAN and port security, and hardens the uplink trunk.

```
ACC-SW01(config)# vlan 10
ACC-SW01(config-vlan)#  name OFFICE_DATA
ACC-SW01(config-vlan)# vlan 20
ACC-SW01(config-vlan)#  name VOICE
ACC-SW01(config-vlan)# vlan 888
ACC-SW01(config-vlan)#  name NATIVE_UNUSED
ACC-SW01(config)# interface range GigabitEthernet1/0/1 - 40
ACC-SW01(config-if-range)#  switchport mode access
ACC-SW01(config-if-range)#  switchport access vlan 10
ACC-SW01(config-if-range)#  switchport voice vlan 20
ACC-SW01(config-if-range)#  switchport port-security
ACC-SW01(config-if-range)#  switchport port-security maximum 3
ACC-SW01(config-if-range)#  switchport port-security mac-address sticky
ACC-SW01(config-if-range)#  switchport port-security violation restrict
ACC-SW01(config-if-range)#  spanning-tree portfast
ACC-SW01(config-if-range)#  spanning-tree bpduguard enable
ACC-SW01(config)# interface TenGigabitEthernet1/1/1
ACC-SW01(config-if)#  switchport mode trunk
ACC-SW01(config-if)#  switchport trunk native vlan 888
ACC-SW01(config-if)#  switchport trunk allowed vlan 10,20,999
```

```
ACC-SW01(config-if)#  switchport nonegotiate
ACC-SW01# copy running-config startup-config
```

Listing 8.2 Cisco IOS XE VLAN and Port-Security Configuration

8.3 PowerShell Script for Fetching Active Directory User Details

The script lists users in an OU with their logon and password details and flags accounts inactive for more than 90 days.

```
param([Parameter(Mandatory)][string]$OU, [int]$InactiveDays = 90)
Import-Module ActiveDirectory -ErrorAction Stop
$cutoff = (Get-Date).AddDays(-$InactiveDays)
$props  = 'DisplayName','Department','Enabled','LastLogonDate',
          'PasswordLastSet','PasswordNeverExpires','LockedOut'
$users  = Get-ADUser -SearchBase $OU -Filter * -Properties $props
$report = foreach ($u in $users) {
    [PSCustomObject]@{
        User        = $u.SamAccountName
        Name        = $u.DisplayName
        Enabled     = $u.Enabled
        LockedOut   = $u.LockedOut
        LastLogon   = $u.LastLogonDate
        PwdAgeDays  = if ($u.PasswordLastSet) {
                        [int]((Get-Date) - $u.PasswordLastSet).TotalDays
                      } else { 'Never' }
        Flag        = if ($u.Enabled -and $u.LastLogonDate -lt $cutoff) {
                        "INACTIVE > $InactiveDays d" } else { '' }
    }
}
$report | Export-Csv ".\ADUserReport.csv" -NoTypeInformation
Write-Host "$($report.Count) users, $(($report | ? Flag).Count) flagged"
```

Listing 8.3 Get-ADUserReport.ps1

CHAPTER 9
SAMPLE OUTPUT (UI WIREFRAMES IN TEXT)

The screens below are text wireframes recreated from observation; all names and values are illustrative.

9.1 Infrastructure Health Monitoring Dashboard

```
+=============================================================================+
| PRTG / Nagios  INFRASTRUCTURE OVERVIEW                      [Refresh: 60s] |
| SENSORS:  [ UP 412 ]  [ WARNING 6 ]  [ DOWN 1 ]  [ PAUSED 3 ]             |
+-------------------------+---------------------------------------------------+
| DEVICE TREE             |  CPU LOAD - sap-prd-db01 - LAST 24 HOURS (%)      |
| v SAP Landscape         |   80|- - - - - - - - - - - - - - - - - - warning  - -|
|   [UP] sap-prd-db01     |   60|                          ####                 |
|   [UP] sap-prd-app01    |   40|       ##     ####    ########      ###       |
|   [WN] sap-qas-db01     |   20| ######################################  #  |
| v Infrastructure        |    0+----+-----+-----+-----+-----+-----+-----+----+|
|   [UP] ad-dc01          |     00   03    06    09    12    15    18    21    |
|   [DN] print-srv02      +---------------------------------------------------+
| v Network               |  Host            Uptime        Avail.   Status     |
|   [UP] core-sw01        |  sap-prd-db01    41d 06h 12m   100.00%  [UP]       |
|   [UP] fw-cluster       |  ad-dc01         88d 14h 03m   100.00%  [UP]       |
|                         |  print-srv02      0d 00h 00m    97.41%  [DOWN]     |
+-------------------------+---------------------------------------------------+
| ALERTS: 09:12 DOWN print-srv02 Ping 100% loss [Ticket] | 08:55 WARN hana/log |
+=============================================================================+
```

Fig. 9.1 Infrastructure Health Monitoring Dashboard

9.2 SAP S/4HANA Transaction Screen (MMBE)

```
+=============================================================================+
| Menu  Edit  Goto  System  Help                                  SAP  _ [] X |
| [/n MMBE        ] [v]   (Save) (Back) (Exit) (Cancel)   (Find)   (?)        |
+-----------------------------------------------------------------------------+
|  Stock Overview: Company Code/Plant/Storage Location/Batch       (Execute F8)|
|    Material          [ 100045872  ]      Plant  [ 1100 ]  BHEL Trichy - HPBP |
+-----------------------------------------------------------------------------+
|  Client/Company Code/Plant/SLoc  | Unrestricted | Quality Insp. | Blocked    |
|  --------------------------------+--------------+---------------+----------- |
|  Full Client                     |   18,450.000 |     2,300.000 |   150.000  |
|    1100 HPBP Trichy              |   12,200.000 |     2,300.000 |   150.000  |
|      0001 Main Stores            |   10,900.000 |     2,300.000 |     0.000  |
|    1200 SSTP Trichy              |    6,250.000 |         0.000 |     0.000  |
+-----------------------------------------------------------------------------+
| [v] PRD (1) 100  | sap-prd-app01 | INS                          09:16:42 |
+=============================================================================+
```

Fig. 9.2 SAP GUI Stock Overview (MMBE)

9.3 ITSM Service Desk Incident Form

```
+=============================================================================+
| servicenow  > Incidents > INC0012847                    [Update] [Resolve]  |
+-----------------------------------------------------------------------------+
|  Caller   [ Stores Supervisor, HPBP ]     Config. item [ sap-prd-app01     ] |
|  Category [ Software / SAP Output   ]     State        [ In Progress     v ] |
|  Impact   [ 2 - Medium ]  Urgency [ 1 - High ]  Priority [ P2 - High (calc)] |
|  Assignment group [ DTG-SAP-BASIS ]   Assigned to [ SAP Basis Engineer 2 ]   |
|  Short description [ GR slips from MIGO not printing at HPBP stores        ] |
+-----------------------------------------------------------------------------+
|  TASK SLAs            Stage         Elapsed   Target   % Elapsed             |
|  P2 Response          Achieved      00:11     00:30    [####------]  37%     |
|  P2 Resolution        In progress   02:24     08:00    [###-------]  30%     |
```

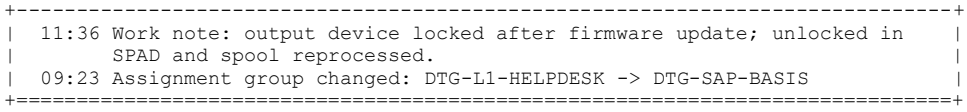


Fig. 9.3 ServiceNow Incident Form with SLA Timers

CHAPTER 10
ANALYSIS

10.1 Security Analysis

Isolating OT from IT through a DMZ removes direct attack paths into systems whose compromise has physical consequences. SCADA servers and PLCs often use protocols such as Modbus/TCP that have no authentication, while enterprise IT is constantly exposed to e-mail, web browsing and removable media. Because every IT-to-OT session must terminate on a hardened DMZ server behind two firewall tiers, a compromised office PC or stolen credential has no route to the control network.

The DMZ also contains malware: ransomware that spreads over SMB or RDP cannot reach across the boundary, and OT updates are staged on DMZ relay servers instead of being downloaded from the internet. Finally, all cross-boundary traffic passes through few enforcement points, so every flow and every vendor session through the MFA jump host is logged and attributable, in line with the zone-and-conduit approach of IEC 62443.

10.2 Storage Analysis

SAN delivers higher IOPS at lower latency than NAS because of its shorter I/O path: a block request goes from the host through the multipath driver and HBA, across a dedicated lossless Fibre Channel fabric, directly to the array cache, with no file-system processing at the storage end. NAS requests pass through the NFS or SMB stack, TCP/IP and a shared Ethernet network, and the NAS must also handle metadata operations such as lookups, permissions and locks. SAN therefore suits latency-critical single-owner volumes such as the HANA redo log, while NAS suits shared files, as summarised in Table 10.1.

Metric	SAN (FC / NVMe-FC, all-flash)	NAS (NFS / SMB over Ethernet)
Small random I/O latency	About 0.1–0.5 ms	About 0.5–3 ms
Random 4–8 KB IOPS	Hundreds of thousands and above	Tens to low hundreds of thousands
Metadata overhead	None at the storage end	Significant (lookup, attributes, locks)

Metric	SAN (FC / NVMe-FC, all-flash)	NAS (NFS / SMB over Ethernet)
Best-suited workloads	HANA data and log, VM datastores	File shares, documents, transport directory

Table 10.1 Indicative SAN vs NAS Performance

CHAPTER 11
CONCLUSION

The summer internship in the Digital Transformation Group of BHEL Tiruchirappalli gave a structured, practical view of enterprise IT inside a large public sector manufacturer. Over thirty days, the work progressed from campus networking and directory services through Linux administration, the PowerVM and SAN-based data centre and the SAP S/4HANA landscape to Check Point security and ITIL service management, showing how each layer depends on the others.

The most important outcome was turning academic knowledge into working skill. Subnetting became an address plan with capacity and security consequences, operating-system concepts became Bash scripts and log analysis, database theory became an understanding of in-memory HANA and replication, and network security became firewall rule reviews and DMZ design. Writing a monitoring-ready health-check script, configuring VLANs and port security, producing an Active Directory access report and tracing an order through MM, PP, QM, SD and FICO each made an abstract topic concrete.

Equally valuable was the discipline of enterprise execution: every change followed an RFC, CAB approval, a maintenance window and a rollback plan, every SAP change moved through DEV, QAS and PRD, and every request was logged and measured against an SLA. The internship showed that reliability comes from consistent process as much as from technology, and it has given a strong foundation for a career in enterprise IT.

CHAPTER 12
FUTURE WORK

- **IoT integration with SAP:** Machine sensor data passed through the IT–OT DMZ could automatically create Plant Maintenance notifications, enabling predictive maintenance and real-time work-centre status in Production Planning.
- **Zero-Trust network architecture:** Every access request would be verified by identity, device posture and context, with micro-segmentation of data-centre workloads and conditional access for SAP and vendor sessions.
- **AI in IT service management:** AI could classify and route tickets, correlate monitoring alerts into single probable-cause incidents and let virtual agents fulfil routine requests such as password resets.
- **Hybrid cloud migration:** Non-production SAP systems, disaster-recovery capacity and analytics could move to a private or government-empanelled cloud while sensitive production data stays in the captive data centre.
- **Infrastructure automation:** Ansible playbooks could codify server builds, patching and switch configuration, detect drift from approved baselines and link standard changes to ServiceNow records.

CHAPTER 13
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